

Proposed Solution

Date	27 JUNE 2026 P
Team ID	XXXX
Project Title	Visualization Tool for Electric Vehicle Charge
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

Electric Vehicles (EVs) are becoming increasingly popular because they reduce pollution and fuel costs. However, many EV users face challenges such as battery range limitations, finding charging stations, and monitoring battery performance. This project provides a visualization tool that helps users track battery charge, estimate driving range, and locate nearby charging stations through interactive dashboards.

Project Overview	
Objective	The main objective of the Visualization Tool for Electric Vehicle Charge and Range Analysis is to help Electric Vehicle (EV) users monitor their battery status, analyze driving range, and make better charging decisions through an interactive visualization dashboard.
Scope	The Visualization Tool for Electric Vehicle Charge and Range Analysis is designed to help EV users monitor battery performance and improve charging efficiency through data visualization. Scope
Problem Statement	
Description	The increasing adoption of Electric Vehicles (EVs) has created a need for efficient battery monitoring and charge management. Many EV users face challenges such as limited battery range, difficulty in locating nearby charging stations, and a lack of clear visualization of battery performance. This often leads to range anxiety, inefficient trip planning, and unnecessary charging delays.

Impact	The Visualization Tool for Electric Vehicle Charge and Range Analysis improves the overall EV user experience by providing clear and interactive insights into battery performance and charging activities.
Proposed Solution	
Approach	The Visualization Tool for Electric Vehicle Charge and Range Analysis follows a simple and systematic approach to monitor battery performance and present useful insights through interactive visualizations.
Key Features	The Visualization Tool for Electric Vehicle Charge and Range Analysis includes the following key features: 🔋 Real-Time Battery Monitoring – Displays the current battery charge level of the EV. 📊 Interactive Dashboard – Visualizes battery status, charging history, and performance using charts and graphs. 🚗 Range Prediction – Estimates the remaining driving distance based on the current battery level. -

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU	Intel i5 or higher / T4 GPU
Memory	RAM	8 GB
Storage	Disk Space	256 GB SSD Or Above
Software		
Frameworks	Backend Framework	Flask
Libraries	Python Libraries	scikit-learn, pandas, numpy, matplotlib,seaborn
Development Environment	IDE	Jupyter Notebook / pycharm / VS Code

Data		
Data	Source, size, format	User Health & Lifestyle dataset (CSV) ,Public Well-Being datasets